

Fabrication of $\text{Li}_2\text{O}-\text{B}_2\text{O}_3-\text{P}_2\text{O}_5$ solid electrolyte by aerosol flame deposition for thin film batteries

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Abstract

Amorphous $\text{Li}_2\text{O}-\text{B}_2\text{O}_3-\text{P}_2\text{O}_5$ thin films were prepared from an aqueous precursor solution of a LiNO_3 and BCl_3 , POCl_3 by Aerosol Flame Deposition. The aqueous precursor solution of the LiNO_3 was first atomized with an ultrasonic vibrator (1.7 MHz), and B_2O_3 and P_2O_5 were synthesized from BCl_3 and POCl_3 by an oxygen–hydrogen flame. Their microstructure and electrical properties were investigated by SEM, XRD, Raman spectroscopy and impedance spectroscopy.

XRD analysis revealed that a crystalline phase of LiCl was formed in the glass soot as well as amorphous boron and phosphorus oxides, and the crystalline LiCl found in deposited glass soot completely disappeared by heat treatment. The correlation between the structural modification of glass network, compositional variation and the conductivity was characterized by Raman spectroscopy. The glass electrolyte fabricated in this study exhibited a maximum conductivity of about 10^{-7} S/cm at 30 sccm flow rate of BCl_3 and POCl_3 .

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1. Introduction

Small, light-weight rechargeable batteries with reasonable capacity are becoming a major issue in the market for small electronic devices, such as active RF cards, micro robots and meso-machines (MEMS) as well as already commercialized consumer electronics. To realize thin film batteries, a solid electrolyte with electrochemical stability and ion mobility is indispensable and recently many researchers have tried to develop the appropriate materials and fabrication processes for solid electrolytes of various types.

Solid electrolytes for lithium batteries are generally divided into organic polymer-based and inorganic-based ones, although the former is not common in thin film-type electrolytes. Inorganic electrolytes may be classified into non-oxide or oxide in terms of material used, and amorphous or crystalline in terms of material structure [1–5]. Amorphous inorganic glass electrolytes have several merits over its crystalline counterparts such as isotropic electrical properties, the ease of preparation,

and the wide choice of composition ensuring a continuous control of electrical properties. Glass electrolytes may be divided into non-oxide (mainly sulfide and oxynitride) and oxide. In spite of high-ion conductivity, the sulfide glasses have several drawbacks such as corrosive chemical nature of sulfur resulting in the difficulty in handling and managing the fabrication, low decomposition voltage and highly hygroscopic nature, and so on. Due to these problems sulfide glass electrolytes have not yet been developed for real application. LiPON electrolytes demonstrated the successful performance as a candidate for thin film electrolytes, although for fabrication it requires plasma under vacuum.

Compared to these materials, the oxide glasses have advantages such as the inertness in atmosphere, wide range of composition to choose, and easy fabrication of thin films. However, the oxide glasses have serious problems of low Li ion conductivity. This drawback of oxide glasses can be less serious if these materials are employed in thin or thick film-type rechargeable batteries, where the internal resistance is reduced due to extremely small thickness.

In this study, we introduce a completely new method for the fabrication of the oxide glass electrolyte for thin film batteries,

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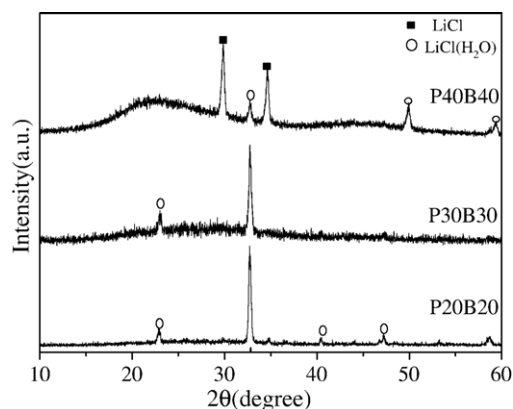


Fig. 1. X-ray diffraction patterns of particles produced at various flow rates of BCl_3 and POCl_3 .

which is called “Aerosol Flame Deposition” (AFD) method. A similar method has been employed in forming porous and spherical oxide particles, especially in optical fiber industry. The synthesized particles are deposited on a substrate in the form of a porous film and consolidated to a film by proper heat treatment. In this work, the Aerosol Flame Deposition was applied to fabricate glass electrolyte film of $\text{Li}_2\text{O}-\text{B}_2\text{O}_3-\text{P}_2\text{O}_5$ and this glass electrolyte was analyzed for material properties such as microstructure/morphology and the basic electrical properties.

2. Experiment

The precursor solution of lithium was prepared by dissolving lithium nitrate in methanol at concentration of 0.5 mol% and it was nebulized with an ultrasonic resonator (1.7 MHz). The aerosol generated by the nebulizer was subsequently carried into the central tube of the torch by flowing dry Ar gas. The vapors of POCl_3 and BCl_3 produced by Ar bubbling under controlled temperature were delivered to an oxygen–hydrogen torch and hydrolyzed to form a porous oxide soot film on Si substrates. The flow rates of source gases were 20, 30, and 40 sccm/min, respectively. The notations for the prepared specimen are the abbreviation of the flow rate of these source gases for the glass formers. For example, B20P20 denotes the specimen prepared at the flow rate of 20 sccm of BCl_3 and 20 sccm of POCl_3 .

To deposit thin films of uniform thickness, glass soot was first deposited on a silicon or quartz glass substrate placed in a rotating stage, which was kept at 150 °C to eliminate H_2O produced during the fuel reaction of hydrogen and oxygen. LiNO_3 , POCl_3 and BCl_3 were hydrolyzed in an oxygen–hydrogen flame and converted into oxide or hydroxide by hydrolysis reaction and grown to particles by nucleation and subsequent Brownian coagulation. The deposited film was sintered into a dense film in a furnace at 700 °C for 10 h.

X-ray diffraction analysis was performed to confirm the formation of lithium related compound particles and to check the crystalline phases of the deposited films. The correlation between the structure and the composition of the electrolyte produced by heat treatment was investigated by Raman analysis

NRS-3100 JSACO with 514 nm argon laser. To carry out the various measurements, soot specimen was used, as well as films. Dense bulk specimen was prepared from the soot collected from the deposited films by melting in a Pt crucible at 1000 °C.

To measure the ionic conductivities, silver paste was painted as electrodes on both faces of the polished specimens. The diameter and thickness of the glass electrolyte specimen were 0.2 mm and 0.8 mm. Conductivity was measured by complex impedance spectroscopy using Solartron 1260 impedance analyzer and dielectric interface with an ac voltage of 50 mV amplitude over the frequency range 1 Hz to 1 MHz at room temperature.

3. Results and discussions

The glass soot was deposited at an oxygen flow rate of 7.5 L/min, Ar shield flow rate of 2.5 L/min and hydrogen flow rate of 1.5 L/min. The supplied source gases of BCl_3 and POCl_3 were transformed into amorphous oxide particles as shown in Fig. 1 [6]. Amorphous patterns were increased with increasing flow rate of BCl_3 and POCl_3 , which suggests that the deposition rate of amorphous oxides increased with the flow rate. Crystalline peaks observed in XRD patterns were assigned to LiCl or LiCl hydrate, which were the reaction products between Li ion from the Li source and Cl ion decomposed from BCl_3 or POCl_3 . The hygroscopic nature of synthesized oxides and LiCl generated a physical and chemical adsorption of moisture to soot particles during deposition or handling in air after deposition. The formed hydroxides or hydrates, however, were not a serious problem since those are eventually converted into oxide during the high temperature heat treatment consolidating the film. After heat treatment at 1000 °C, X-ray diffraction patterns of glass soot collected from the deposited films were summarized in Fig. 2. One can see that the crystalline LiCl(H_2O) and LiCl found in glass soot completely disappeared by heat treatment.

Fig. 3(a) shows Nyquist plots of the $\text{Li}_2\text{O}-\text{B}_2\text{O}_3-\text{P}_2\text{O}_5$ glass electrolyte at various flow rates of BCl_3 and POCl_3 , and the conductivity dependency on the concentration of network former, $\text{B}_2\text{O}_3-\text{P}_2\text{O}_5$, is shown in Fig. 3(b). The lithium-ion conductivity is determined mainly by the concentration of

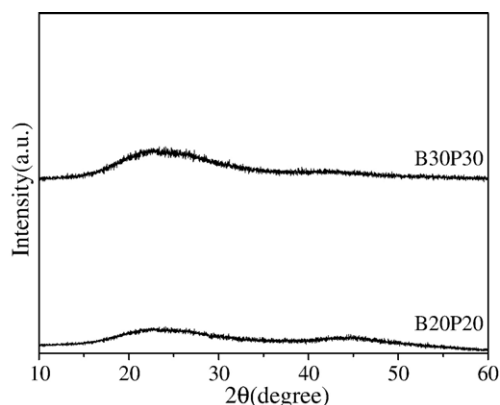


Fig. 2. XRD patterns of glass prepared by melting at 1000 °C.

lithium ions in the glass network and the variation of conductivity shown in Fig. 3(b) reflects the effective lithium-ion concentration controlled by changing the deposition rate of the glass formers. With increasing flow rate of BCl_3 and POCl_3 , the amount of glass formers would increase and, thus, the effective concentration of lithium ion in the synthesized soot would decrease since the flow rate of lithium source was fixed. The lithium-ion conductivity of the glass specimen B20P20 decreased even though the concentration of lithium incorporated into the glass was higher than other specimen. The cause of this abnormal result is believed to be due to the micro-scale segregation of lithium ions or the formation of an amorphous Li_2O phase. The micro-scale segregation is a phenomenon normally observed in glasses with high concentration of alkali ions, which is similar to phase separation with an extremely mild concentration gradient in molecular level. The existence of this phenomenon in the prepared specimen is supported by the XRD pattern shown in Fig. 2. The broad amorphous peaks appeared at two different angles and this suggests that mild phase separation or segregation occurred. The amorphous peaks are more clearly separated in the specimen B20P20, which suggests that the possibility of micro-segregation is higher for

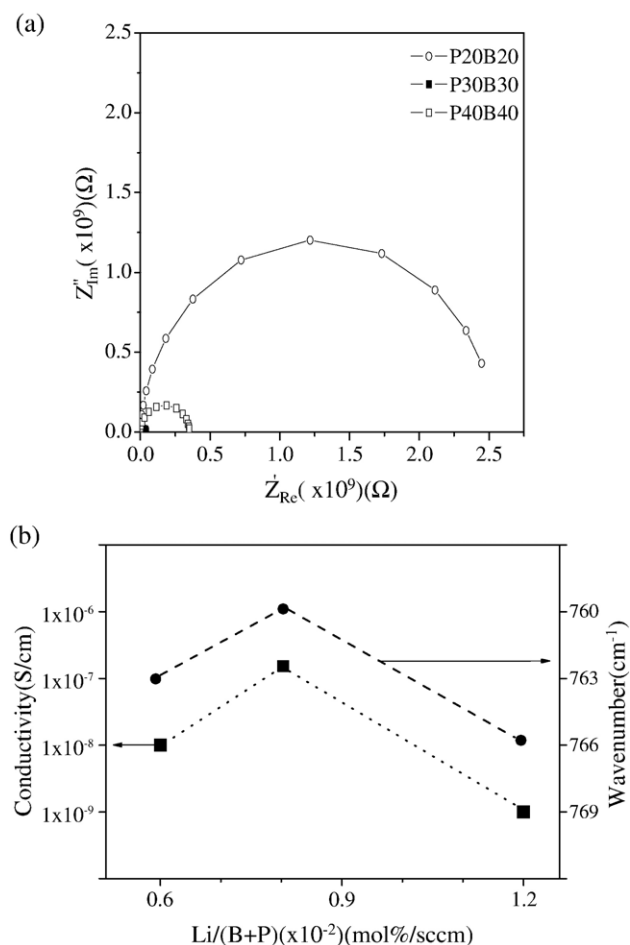


Fig. 3. (a) Nyquist plots of bulk glass electrolyte fabricated from soot collected at various flow rates of BCl_3 and POCl_3 , and (b) the measured conductivity and the peak position of diborate unit as functions of $\text{Li}/(\text{B}+\text{P})$ ratio.

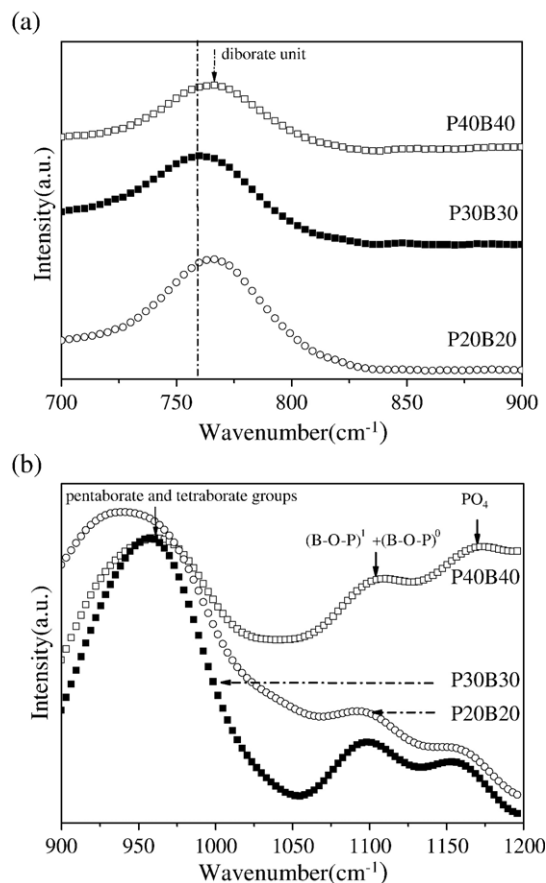


Fig. 4. Raman spectra of glass electrolyte at various flow rate of BCl_3 and POCl_3 . (a) The diborate unit $(\text{B}_4\text{O}_7)^{2-}$, and (b) the band between 1100 cm^{-1} and 1170 cm^{-1} .

this glass. This micro-segregation would reduce the lithium conductivity by diminishing the mobile lithium ions loosely attached to the nonbridging oxygens as experimentally observed.

The relationship between structure and conductivity is investigated by Raman spectra. The observed Raman spectra were similar to that of pure borate glass with lithium ions and this indicates that the glass prepared in this work is almost pure

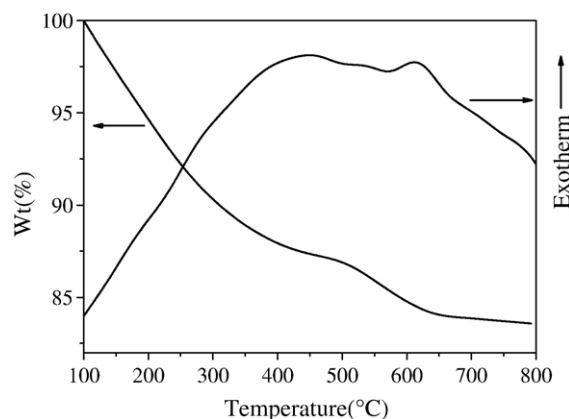


Fig. 5. TGA–DSC curves obtained from collected glass soot synthesized at the flow rate BCl_3 : 30 sccm, POCl_3 : 30 sccm and Li : 0.5 mol%.

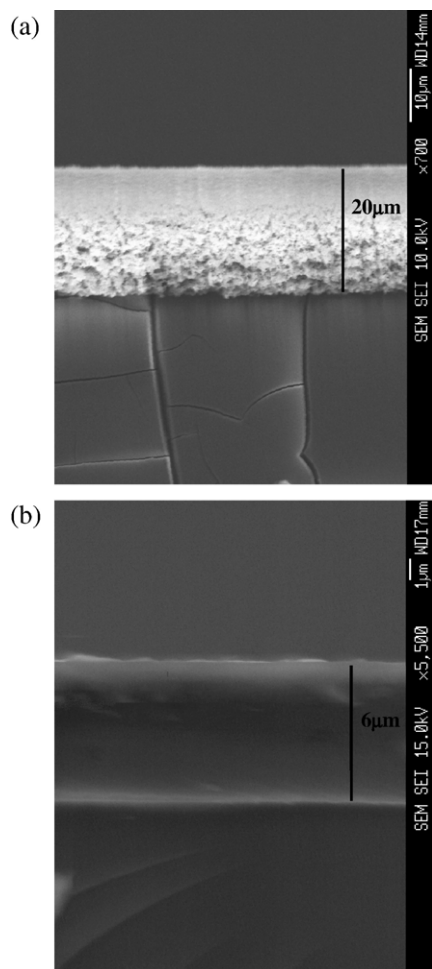


Fig. 6. Cross-sectional images of deposited glass soot and sintered film. (a) Glass soot synthesized at flow rate BCl_3 : 30 sccm, POCl_3 : 30 sccm and Li: 0.5 mol% on silicon wafer before heat treatment, and (b) thin film obtained by heating the deposited soot film at 700 °C for 10 h.

borate glass. This is due to the low synthesis rate of P_2O_5 compared to B_2O_3 even under the same flow rate of both source gases and intense evaporation of P_2O_5 due to its highly volatile nature. Nevertheless, P_2O_5 is observed to exist in the specimen since the peak at 1170 cm^{-1} , PO_4 , was still detected as shown in Fig. 4(b). The Raman spectra of pure B_2O_3 was observed at 806 cm^{-1} assigned to the vibration of B–O bonding in a boroxol ring without nonbridging oxygen, and reported to shift to lower wave numbers and to broaden with increasing Li concentration [7–9]. The shift to lower wave numbers amounts to as much as 40 cm^{-1} . The peak around 760 cm^{-1} observed in this work was assigned to the vibration of diborate groups $(\text{B}_4\text{O}_7)^{2-}$, which is a typical glass network structure with nonbridging oxygen formed by added lithium ion [10]. Namely, the pure boroxol ring converts into diborate by formation of nonbridging oxygens. The vibration of diborate groups $(\text{B}_4\text{O}_7)^{2-}$ itself was observed to shift with changing the flow rate of BCl_3 and POCl_3 as shown in Fig. 4(a). It is obvious that the conductivity follows the variation of peak position of diborate as shown in Fig. 3(b) and this variation is ascribed to the variation of the concentration of “mobile” lithium ions. The peak at 950 cm^{-1} also exhibited a similar behavior to 760 cm^{-1} peak as

shown in Fig. 4(b). The peak at 1110 cm^{-1} , which is assigned to the combination of various P–O groups with $(\text{B–O–P})^{1-}$ and $(\text{B–O–P})^0$, also shifted to lower wavelength for B30P30 specimen, reflecting the conversion of $(\text{B–O–P})^0$ into $(\text{B–O–P})^{1-}$ [11].

The Raman spectra reveal that the peaks generally are very pronounced and sharp suggesting that a large amount of Li is added into the glass network and a large amount of nonbridging oxygen is produced [12]. The amount of Li added into the glass network is believed to reach the limit, $\sim 50\text{ mol}\%$, which glass networks can sustain. The amount of residual P_2O_5 seems to be lower than as expected from the flow rate of the source gas.

TGA–DSC measurements were carried out to find the optimum sintering temperature of glass soot B30P30, and the result is shown in Fig. 5. The TGA curve exhibits a weight loss of 15% in the temperature range from 100 °C to 450 °C. Although the glass soot had been kept in dry atmosphere this phenomenon seems to result from water molecules in the case of the particles containing the crystalline phases. Weight loss was observed in the temperature range from 520 °C to 650 °C due to LiCl, and above 650 °C no weight loss was observed as a function of temperature change. The DSC curve increase to 650 °C, and at this temperature began to decrease. The glass soot began to densify at this temperature.

The cross-section of glass soot deposited by AFD before heat treatment is shown in Fig. 6(a) and the thickness of soot film was about 20 μm . After heat treatment at 700 °C, a dense film of glass was produced, as shown in Fig. 6(b). The deposition rate estimated after the consolidation was 0.3 $\mu\text{m}/\text{min}$, which is much faster than with common thin film deposition equipment such as sputtering.

4. Conclusions

The inorganic glass electrolytes $\text{Li}_2\text{O–B}_2\text{O}_3\text{–P}_2\text{O}_5$ were deposited by Aerosol Flame Deposition at various flow rates of POCl_3 and BCl_3 . The soot film contained crystalline LiCl and this phase was completely converted into an amorphous oxide by high temperature heat treatment. The glass electrolytes produced at a flow rate of 30 sccm of POCl_3 and BCl_3 had the highest conductivity of 10^{-7} S/cm at room temperature. This conductivity is comparable to that of LiPON. Dense film of 6 μm thickness was obtained with a maximum deposition rate of 0.3 $\mu\text{m}/\text{min}$. It was demonstrated that Aerosol Flame Deposition is a good candidate for the fabrication of thin film glass electrolyte.

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